THE PUSHFORWARD-PULLBACK COMPOSITE OF A FINITE LOCALLY FREE MORPHISM IS MULTIPLICATION BY THE DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author's sentence at source lines 2457-2458, 'We omit showing that a finite morphism has relative dimension 0.', is retained verbatim inside the counted proof; the degree computation is inside the span.

Allowed standard prerequisites (editorial).

- A finite locally free morphism is flat and finite, a finite morphism is proper, and finite local freeness is preserved by base change. Morphisms, labels morphisms-lemma-finite-flat (proof line 2454), morphisms-lemma-finite-proper (proof line 2456) and morphisms-lemma-base-change-finite-locally-free (proof line 2463). Gives the properness and flatness assertions and shows that the base change of the morphism over an integral closed subscheme is again finite locally free of the same degree.
- Cycle class of a closed subscheme, flat pullback and proper pushforward of cycle classes. Same chapter chow.tex, labels lemma-pullback-coherent (proof line 2470) and lemma-cycle-push-sheaf (proof line 2470). Reduces the formula for an integral class to the statement that the pushforward of the pulled back structure sheaf is a finite locally free sheaf of the given rank, giving the factor d.

Source reference. Original lemma 02R6: sources/stacks-chow/pinned-original-source.tex, source lines 1992–1992. Navigation pointer for \ref{lemma-cycle-push-sheaf} used

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inside the counted span (line 2471); the argument that consumes it is retained in the span.

Source reference. Original lemma 02RE: sources/stacks-chow/pinned-original-source.tex, source lines 2331–2331. Navigation pointer for \ref{lemma-pullback-coherent} used inside the counted span (line 2470); the argument that consumes it is retained in the span.

Source reference. Original definition 02KA: sources/stacks-chow/tags.txt, source lines 3365–3365. Navigation pointer for \ref{morphisms-definition-finite-locally-free} used inside the counted span (line 2444); the argument that consumes it is retained in the span.

Source reference. Original lemma 02KD: sources/stacks-chow/tags.txt, source lines 3368–3368. Navigation pointer for \ref{morphisms-lemma-base-change-finite-locally-free} used inside the counted span (line 2463); the argument that consumes it is retained in the span.

Source reference. Original lemma 02KB: sources/stacks-chow/tags.txt, source lines 3366–3366. Navigation pointer for \ref{morphisms-lemma-finite-flat} used inside the counted span (line 2454); the argument that consumes it is retained in the span.

Source reference. Original lemma 01WN: sources/stacks-chow/tags.txt, source lines 2488–2488. Navigation pointer for \ref{morphisms-lemma-finite-proper} used inside the counted span (line 2456); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 2440); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X, Y be locally of finite type over S. Let $f : X \to Y$ be a finite locally free morphism of degree d (see Morphisms, Definition 02KA). Then f is both proper and flat of relative dimension 0, and*

$$f_* f^* \alpha = d\alpha$$

for every $\alpha \in Z_k(Y)$.

Proof. A finite locally free morphism is flat and finite by Morphisms, Lemma 02KB, and a finite morphism is proper by Morphisms, Lemma 01WN. We omit showing that a finite morphism has relative dimension 0. Thus the formula makes sense. To prove it, let $Z \subset Y$ be an integral closed subscheme of δ-dimension k. It suffices to prove the formula for $\alpha = [Z]$. Since the base change of a finite locally free morphism is finite locally free (Morphisms, Lemma 02KD) we see that $f_* f^* \mathcal{O}_Z$ is a finite locally free sheaf of rank d on Z. Hence

$$f_* f^*[Z] = f_* f^*[\mathcal{O}_Z]_k = [f_* f^* \mathcal{O}_Z]_k = d[Z]$$

where we have used Lemmas 02RE and 02R6. □